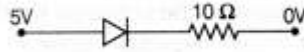
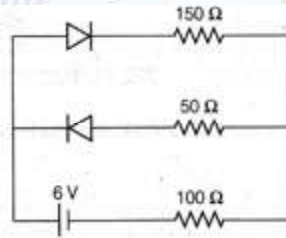


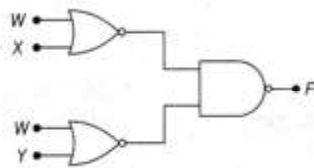
- The temperature coefficient of resistance of a semiconductor
 - is always positive
 - is always negative
 - is zero
 - may be positive or negative or zero
- A junction diode has a resistance of 25Ω when forward biased and 2500Ω when reverse biased. The current in the diode, for the arrangement shown will be



- $\frac{1}{15}\text{A}$
 - $\frac{1}{7}\text{A}$
 - $\frac{1}{25}\text{A}$
 - $\frac{1}{480}\text{A}$
- The circuit shown in the figure contains two diodes each with a forward resistance of 50Ω and with infinite backward resistance. If the battery is 6V, the current through the 100Ω resistance (in ampere) is :



- Zero
 - 0.02
 - 0.03
 - 0.036
- Which of the following logic gate is a universal use of this gate
 - OR
 - NOT
 - AND
 - NOR
 - The diagram of a logic circuit is given below.



The output F of the circuit is given by

- $W \cdot (X + Y)$
 - $W \cdot (X \cdot Y)$
 - $W + (X \cdot Y)$
 - $W + (X + Y)$
- Masses of nucleus, neutron and proton are M , m_n , and m_p , respectively. If nucleus has been divided into neutrons and protons, then
 - $M = (A - Z)m_n + Zm_p$
 - $M = Zm_n + (A - Z)m_p$
 - $M < (A - Z)m_n + Zm_p$
 - $M > (A - Z)m_n + Zm_p$
 - A nucleus with $Z = 92$ emits the following in a sequence.

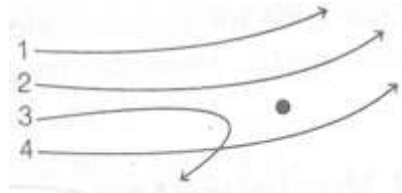
$\alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$. The Z of the resulting nucleus is

- (1) 74 (2) 76 (3) 78 (4) 8

8. Consider α -particle, β -particles and γ -rays, each having an energy of 0.5 MeV. In increasing order of penetrating powers, the radiations are

- (1) α, β, γ (2) α, γ, β (3) β, α, γ (4) γ, β, α

9. The diagram shows the paths of four α -particles, having the same energy, being scattered by the nucleus of an atom simultaneously. Which of these is/are not possible physically?



- (1) 3 and 4 (2) 2 and 3 (3) 1 and 4 (4) 4 only

10. Of the following transitions in hydrogen atom, the one which gives an absorption line of highest frequency is

- (1) $n = 1$ to $n = 2$ (2) $n = 3$ to $n = 8$ (3) $n = 2$ to $n = \infty$ (4) $n = 8$ to $n = 3$

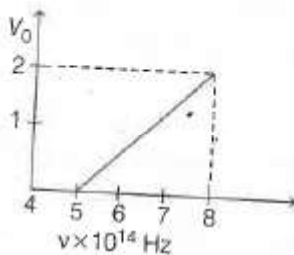
11. Let the potential energy of a hydrogen atom in the ground state be zero. Then, its energy in the first excited state will be

- (1) 10.2 eV (2) 13.6 eV (3) 23.8 eV (4) 27.2 eV

12. If the frequency of light in photoelectric experiment is doubled, the stopping potential will

- (1) be doubled (2) be halved
(3) become more than double (4) become less than double

13. The stopping potential (V_0) versus frequency (ν) plot of a substance is shown in figure, the threshold wavelength is



- (1) 5×10^{14} m (2) 6000 Å
(3) 5000 Å (4) cannot be estimated from given data

14. When stopping potential is applied in an experiment on photoelectric effect, no photocurrent

is observed. This means that

- (1) the emission of photoelectrons is stopped
- (2) the photoelectrons are emitted but are reabsorbed by the emitter metal
- (3) the photoelectrons are accumulated near the collector plate
- (4) the photoelectrons are dispersed from the side the apparatus

15. Two closely located, monochromatic coherent sources create a fixed interference pattern on a screen. The frequency of one of the sources is then decreased. What will happen to the interference pattern as a result?

- (1) The pattern will completely disappear
- (2) The fringe spacing will increase
- (3) The fringes will shift up or down, but the spacing will remain constant
- (4) the fringe spacing will decrease

16. At a certain point on a screen in a YDSE apparatus, the path difference for the two interfering rays is $(1/3)$ rd of the wavelength. Find the ratio of the intensity at this point to that at the centre of a bright fringe.

- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$ (3) $\frac{1}{3}$ (4) $\frac{3}{4}$

17. If the Young's double slit experiment is performed with white light, then which of the following is not true?

- (1) the central maximum will be white
- (2) there will not be a completely dark fringe
- (3) the bright fringe next to the central will be red
- (4) the bright fringe next to the central will be violet

18. Light of wavelength $5000 \text{ \AA}$ incidents over a slit of width $1 \text{ }\mu\text{m}$. The angular width of central maxima will be

- (1) 30° (2) 60° (3) 90° (4) 120°

19. Direction of the first secondary maxima in the Fraunhofer diffraction pattern at a single slit is given by (a is the width of the slit)

- (1) $a \sin \theta = \frac{\lambda}{2}$ (2) $a \cos \theta = \frac{3\lambda}{2}$ (3) $a \sin \theta = \lambda$ (4) $a \sin \theta = \frac{3\lambda}{2}$

20. To observe diffraction, the size of an obstacle

- (1) should be of the same order as wavelength
- (2) should be much larger than the wavelength
- (3) have no relation to wavelength

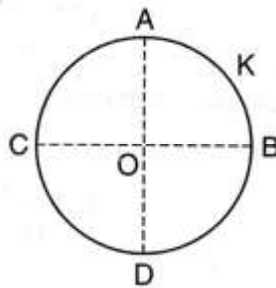
- (4) should be exactly $\frac{\lambda}{2}$

21. An unpolarised beam of intensity I_0 is incident on a pair of nicol prisms making an angle of 60° with each other. The intensity of light emerging from the pair is
- (1) I_0 (2) $\frac{I_0}{2}$ (3) $\frac{I_0}{4}$ (4) $\frac{I_0}{8}$
22. In a single slit, diffraction pattern, as the width of the slit is increased.
- (1) the peak intensity of central maxima increases and its width also increases
(2) the peak intensity of central maxima increases and its width also decreases
(3) the peak intensity of central maxima decreases and its width also increases
(4) the peak intensity of central maxima decreases and its width also decreases
23. The angle of polarisation for any medium is 60° what will be critical angle for this?
- (1) $\sin^{-1} \sqrt{3}$ (2) $\tan^{-1} \sqrt{3}$ (3) $\cos^{-1} \sqrt{3}$ (4) $\sin^{-1} \frac{1}{\sqrt{3}}$
24. The angle of incidence at which reflected light is totally polarised for reflection from air to glass (reflection index n) is
- (1) $\sin^{-1}(n)$ (2) $\sin^{-1}\left(\frac{1}{n}\right)$ (3) $\tan^{-1}\left(\frac{1}{n}\right)$ (4) $\tan^{-1}(n)$
25. In the propagation of electromagnetic waves, the angle between the direction of propagation and plane of polarisation is
- (1) 0° (2) 45° (3) 90° (4) 180°
26. Light waves can be polarised as they are
- (1) transverse (2) of high frequency (3) longitudinal (4) reflected
27. When a polaroid sheet is rotated between two crossed polaroids then the transmitted light intensity will be maximum for a rotation of
- (1) 45° (2) 60° (3) 90° (4) 30°
28. The angle of incidence i for an equilateral prism of refractive index $\sqrt{3}$ so that the ray is parallel to the base inside the prism is
- (1) 30° (2) 45° (3) 60° (4) either 30° or 60°
29. A thin symmetric doubleconvex lens of power P is cut into three parts A, B and C as shown. The power of

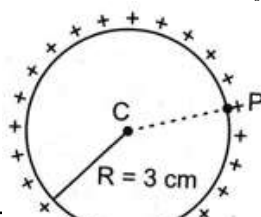


- (1) A is P (2) A is $2P$ (3) B is $2P$ (4) B is $\frac{P}{4}$

30. The focal length of a lens of refractive index $3/2$ is 10 cm in air. The focal length of that lens in a medium of refractive index $7/5$ is
 (1) -70 cm (2) $10/7$ cm (3) 70 cm (4) None of these
31. For a diverging lens that has focal length of magnitude f , the image size was identical to that of the object. How is this possible?
 (1) The object was virtual and located a distance $2f$ from the lens.
 (2) The object was virtual and located a distance f from the lens.
 (3) The object was real and located a distance $2f$ from the lens.
 (4) It is not possible
32. What is the flux through a cube of side ' a ' if a point charge of q is at one of its corner:
 (1) $\frac{2q}{\epsilon_0}$ (2) $\frac{q}{8\epsilon_0}$ (3) $\frac{q}{\epsilon_0}$ (4) $\frac{q}{2\epsilon_0}$
33. The electric field at a distance $\frac{3R}{2}$ from the centre of a charged conducting spherical shell of radius R is E . The electric field at a distance $\frac{R}{2}$ from the centre of sphere is :
 (1) zero (2) E (3) $\frac{E}{2}$ (4) $\frac{E}{3}$
34. A thin conducting ring of the radius R is given charge $+Q$. the electric field at the centre O of the ring due to the charge on the part AKB of the ring is E . The electric field at the centre due to the charge on the part $ACDB$ of the ring is:



- (1) E along KO (2) E along OK (3) $3E$ along KO (4) $3E$ along OK
35. A thin spherical shell is charged by some source. The potential difference between the two point C and P (in V) shown in the figure is: ($\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI units)

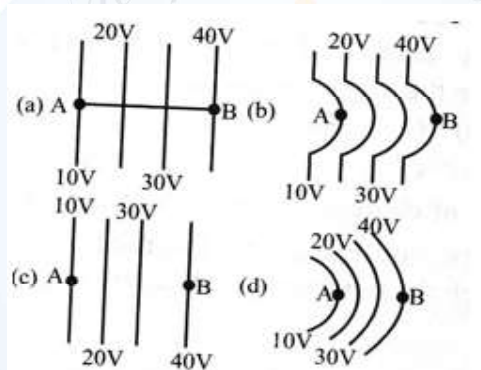


- (1) 0.5×10^5 (2) zero (3) 3×10^5 (4) 1×10^5

36. The angle between the electric lines of force and the equipotential surface is:
(1) 45° (2) 90° (3) 100° (4) 0°

37. Twenty seven drops of same size are charged at 200 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.
(1) 1800 V (2) 660 V (3) 1320 V (4) 1520 V

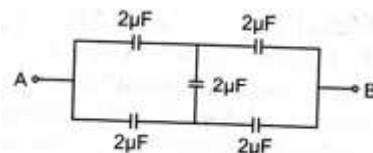
38. The diagrams below show regions of equipotentials.



A positive charge is moved from A to B in each diagram

- (1) In all the four cases the work done is the same
(2) Minimum work is required to move q in figure (1)
(3) Minimum work is required to move q in figure (2)
(4) Maximum work is required to move q in figure (3)

39. In the following circuit, the equivalent capacitance between terminal A and terminal B is :



- (1) $0.5 \mu\text{F}$ (2) $4 \mu\text{F}$ (3) $2 \mu\text{F}$ (4) $1 \mu\text{F}$
(2) Minimum work is required to move q in figure (1)
(3) Minimum work is required to move q in figure (2)
(4) Maximum work is required to move q in figure (3)

40. A parallel plate air capacitor has capacity 'C' distance of separation between plate 'd' and potential

difference 'V' is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is:

(1) $\frac{CV^2}{2d}$ (2) $\frac{CV^2}{d}$ (3) $\frac{C^2V^2}{2d}$ (4) $\frac{C^2V^2}{2d^2}$

41. When light propagates through a material medium of relative permittivity ϵ_r , and relative permeability μ_r , the velocity of light, v is given by : (c-velocity of light in vacuum)

(1) $v = \sqrt{\frac{\mu_r}{\epsilon_r}}$ (2) $v = \sqrt{\frac{\epsilon_r}{\mu_r}}$ (3) $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$ (4) $v = c$

42. An em wave is propagating in a medium with a velocity $\vec{v} = \vec{v}_i$. The instantaneous oscillating electric field of this em wave is along + y axis. Then the direction of oscillating magnetic field of the em wave will be along :

(1) -z-direction (2) +z-direction (3) -x-direction (4) -y-direction

43. If λ_v , λ_x and λ_m represent the wavelength of visible Light, X-rays and microwaves then :

(1) $\lambda_m > \lambda_x > \lambda_v$ (2) $\lambda_m > \lambda_v > \lambda_x$ (3) $\lambda_v > \lambda_x > \lambda_m$ (4) $\lambda_v > \lambda_m > \lambda_x$

44. A plano convex lens fits exactly into a plano concave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then the focal length the combination is:

(1) $\frac{R}{2(\mu_1 - \mu_2)}$ (2) $\frac{R}{(\mu_1 - \mu_2)}$ (3) $\frac{2R}{(\mu_2 - \mu_1)}$ (4) $\frac{R}{(\mu_1 + \mu_2)}$

45. When a biconvex lens of glass having refractive index 1.47 is dipped in a liquid, it acts as a plane sheet of glass. This implies that the liquid must have refractive index :

(1) equal to that of glass (2) less than one
(3) greater than that of glass (4) less than that of glass

ANSWER KEY

1	2	3	4	5	6	7	8	9	10
2	2	2	4	3	3	3	1	4	1
11	12	13	14	15	16	17	18	19	20
3	3	2	2	1	2	3	2	4	1
21	22	23	24	25	26	27	28	29	30
2	4	4	4	1	1	1	3	1	3
31	32	33	34	35	36	37	38	39	40
1	2	1	2	2	2	1	1	3	1
41	42	43	44	45					
3	2	2	2	1					